Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 01: Syllabus and Review of Maxwell Equations

Instructor: Zihan Wang

About Me

- Zihan Wang
- B.S. in Space Physics at Peking University, China in 2016
- Ph.D. in Space Physics at the University of Michigan in 2021
- Joined UTA in 2023
- Research Interests:
  - Terrestrial magnetosphere-ionosphere-thermosphere coupling
  - Data analysis of satellites data and ground-based measurement
  - First-principle modeling
  - Machine Learning

About the Course

- EM 2 (PHYS 5313) is the second semester of a two-semester course in electromagnetic theory at the graduate level.
- We will discuss solutions of Maxwell Equations with different boundary conditions and their applications.
  - Propagation of Electromagnetic waves (no source term)
    - Propagation in vacuum and uniform dielectric medium
    - Propagation at a plane interface: reflection, refraction
    - Dispersion
    - Propagation in conductor
    - Applications: waveguides, resonant cavity, optical fiber, radio waves in the atmosphere

About the Course

- Propagation in nonuniform medium: Scattering
- Application: Color of the sky and rainbow

- Electromagnetic radiation (with source term)
  - Antenna
  - Particles
- Special relativity

Logistics

- **Instructor:** Zihan Wang
- **Email Address:** zihan.wang@uta.edu
- **Class Time:** MoWe 2:30 PM - 3:50 PM
- Office Hour: MoWe 1:30-2:30 PM @ CPB 335
- **Place of Class Meetings:** SH 315
- **Textbook:**
- *Classical Electrodynamics*, 3rd Edition, by John David Jackson, published by John Wiley & Sons.
- *Modern Electrodynamics*, 1st Edition, by Andrew Zangwill, published by Cambridge University Press.
- **Notes**: All lecture notes will be posted on CANVAS after class.

Grades (Final)

- Grade contributions:
  - Homework (x10): 40%
  - 1st Midterm: 20%
  - 2nd Midterm: 20%
  - 3rd Midterm : 20%
  - Participation: 5%

The final grade cutoffs are as follows:

A: 90 or greater

B: 80-89

C: 70-79

D: 60-69

F: 59 or lower

- Homework:
  - Late HW will be graded but will have 50% deducted.
  - Discussion is encouraged, but the submitted work must be your own.
  - Explain the reasoning behind your solutions. Credit is given for the methods.

Exams

- All exams are closed-book.
- One page (front and back) of handwritten notes is allowed.
- Notes can contain equations only.
- All exam problems will be from in-class examples or homework problems.

Recommended further reading

- Introduction to Electrodynamics by Griffiths, David J.
- The Feynman Lectures on Physics Vol. II by Matthew Sands, Richard Feynman, and Robert B. Leighton

Questions?

Overview of Chapter 07

1. Maxwell Eq -> Wave Equations
2. Plane-wave solution
3. Properties of plane electromagnetic waves
   - General solution
   - Polarization
4. Reflection and refraction
5. Dispersion in different mediums
6. Non-monochromatic waves

Maxwell Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

No sources

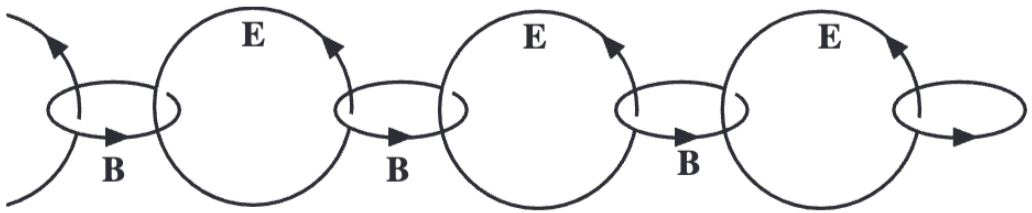
$\vec{J} = 0, \quad \rho = 0$

No boundry

Maxwell Equations in vacuum

No sources (j=0, ρ=0)
No boundary.

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \end{cases}$$



"Mutual embrace" by Maxwell

Wave Equation

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \end{cases} \Rightarrow \quad \nabla \times (\nabla \times \vec{E}) = -\nabla \times \frac{\partial \vec{B}}{\partial t}$$

$$\nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

$$-\nabla^2 \vec{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\Rightarrow \quad \nabla^2 \vec{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \qquad \text{Wave equation}$$

$$\mu_0 \varepsilon_0 = \frac{1}{c^2}$$

Wave Equation

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \end{cases} \Rightarrow$$

$$\nabla \times (\nabla \times \vec{B}) = \mu_0 \varepsilon_0 \nabla \times \frac{\partial \vec{E}}{\partial t}$$

$$\nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \nabla \times \vec{E}$$

$$-\nabla^2 \vec{B} = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\nabla^2 \vec{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

Wave Equation

$$\nabla^2 \mathbf{E} - \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} = 0$$

$$\nabla^2 \mathbf{B} - \frac{1}{c^2}\frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.$$

How to solve the wave equation?

- Start from one component, e.g., Ex
- Fourier integral

$$\nabla^2 \bar{E}_x - \frac{1}{c^2}\frac{\partial^2 \bar{E}_x}{\partial t^2} = 0$$

$$\bar{E}_x(\vec{x}, t) = \frac{1}{(2\pi)^{3/2}} \int d^3k \, e^{i\vec{k}\cdot\vec{x}} \, e_x(\vec{k}, t)$$

$$e_x(\vec{k}, t) = \frac{1}{(2\pi)^{3/2}} \int d^3x \, e^{-i\vec{k}\cdot\vec{x}} \, \bar{E}_x(\vec{x}, t)$$

$$\nabla^2 e^{i\vec{k}\cdot\vec{x}} = -k^2 e^{i\vec{k}\cdot\vec{x}}$$

How to solve the wave equation?

$$\nabla^2 \bar{e}_x = \frac{1}{(2\pi)^{3/2}} \int d^3k \, e_x(\vec{k}, t)(-k^2) e^{i\vec{k}\cdot\vec{x}}$$

$$\| \qquad\qquad \|$$

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\bar{e}_x = \frac{1}{c^2}\frac{1}{(2\pi)^{3/2}} \int d^3k \, \frac{\partial^2 e_x}{\partial t^2} e^{i\vec{k}\cdot\vec{x}}$$

$$\Rightarrow \quad -k^2 e_x = \frac{1}{c^2}\frac{\partial^2 e_x}{\partial t^2} \quad \Rightarrow \quad e_x + \frac{1}{k^2 c^2}\frac{\partial^2 e_x}{\partial t^2} = 0$$

$\omega = kc$ Angular frequency

Helmholtz equation

$$e_x + \frac{1}{k^2 c^2} \frac{\partial^2 e_x}{\partial t^2} = 0$$

$$\frac{\partial^2 e_x}{\partial t^2} + k^2 c^2 e_x = 0$$

$$e_x(\vec{k}, t) = A(\vec{k}) e^{i\omega t} + B(\vec{k}) e^{-i\omega t} \qquad \omega = kc$$

$$\overline{e}_x(\vec{x}, t) = \frac{1}{(2\pi)^{3/2}} \int d^3k \, e^{i\vec{k}\cdot\vec{x}} \left(A(\vec{k}) e^{i\omega t} + B(\vec{k}) e^{-i\omega t} \right)$$

Monochromatic plane wave

$$\vec{E}_x \propto e^{i\vec{k}\cdot\vec{x} \pm i\omega t}$$

$$\vec{E}(\vec{x},t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} \pm i\omega t)}$$

$$\vec{B}(\vec{x},t) = \vec{B}_0 e^{i(\vec{k}\cdot\vec{x} \pm i\omega t)}$$

Monochromatic plane wave

Initial condition: $t=0$, $\bar{b}_x = \alpha(\vec{x})$

$$\frac{\partial \bar{b}_x}{\partial t} = \beta(\vec{x})$$

$\Rightarrow$ $A(\vec{k})$? $B(\vec{k})$?